



CONNECTMEET

Project Documentation

A modern Zoom-style video conferencing web application for real-time online meetings



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01. PROJECT AT A GLANCE

ConnectMeet is a Zoom-style video conferencing web application MVP designed for real-time online meetings.

The application provides users with a centralized interface to create meetings, join existing meetings, schedule meetings, communicate through video and audio, share screens, manage participants and interact through reactions and raise-hand functionality.

The product was first designed as a complete connected UI system in **Build Design Mode** and was then converted into a functional application through **Build Code Mode**.

Core Capabilities

- New Meeting
 - Join Meeting
 - Schedule Meeting
 - Pre-Join Lobby
 - Video Calling
 - Audio Calling
 - Participant Management
 - Screen Sharing
 - Active Speaker Detection
 - Participant Pinning
 - Raise Hand
 - Emoji Reactions
 - Chat
 - Auto Reconnect
 - Meeting Controls
 - Responsive Mobile Experience
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02. PROBLEM, USERS & PRODUCT VALUE

Problem

Remote teams, students, businesses and distributed groups need a simple way to communicate through real-time video meetings.

A conferencing platform needs to handle more than simply displaying video. Users need meeting creation, joining, device permissions, participant controls, screen sharing, reactions, connection recovery and responsive layouts.

ConnectMeet addresses these requirements through a unified video-conferencing experience.

Users

Meeting Host

Creates meetings, manages participants, controls meeting settings and can end meetings for everyone.

Participant

Joins meetings, controls their microphone and camera, shares their screen, reacts, raises their hand and interacts with other participants.

Meeting Organizer

Schedules upcoming meetings and manages meeting information.

Remote Team Member

Uses ConnectMeet for team meetings, collaboration, presentations and communication.

Product Value

- Real-time video communication
 - Easy meeting creation
 - Simple meeting joining
 - Screen sharing
 - Participant management
 - Active speaker detection
 - Reactions and raise hand
 - Automatic reconnection
 - Responsive meeting experience
 - Unified meeting interface
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03. CUSTOMER PERSONA

Primary Persona: Remote Team Member / Meeting Organizer

Goal: Conduct reliable online meetings with multiple participants while having simple controls for communication and collaboration.

Needs

- Quickly create meetings
- Join meetings using an ID or link
- Preview camera and microphone before joining
- Control audio and video
- Share screen
- View participants
- Identify the active speaker
- Raise hand during meetings
- Send reactions
- Recover from unstable connections
- Use meetings across desktop, tablet and mobile

Pain Points

- Complicated meeting interfaces
- Unclear microphone or camera states
- Difficulty managing participants
- Poor experience on mobile devices
- Unstable network connections
- Lack of clear meeting controls
- Difficulty switching between speaker and gallery views

ConnectMeet Solution

ConnectMeet provides a complete meeting workflow starting from meeting creation and joining, followed by a pre-join lobby and real-time meeting interface with participant controls, screen sharing, reactions, active speaker detection and automatic reconnection.

04. ARCHITECTURE & TECHNOLOGY

ConnectMeet is designed as a real-time video conferencing application consisting of a responsive frontend, meeting interface and real-time communication layer.

Frontend

- React.js
- Responsive UI
- Component-based architecture
- Modern UI components
- Responsive meeting layouts

Real-Time Communication

- Peer-to-peer or SFU-based video and audio communication
- Real-time participant state synchronization
- Audio and video track management
- Screen sharing
- Active speaker detection
- Connection monitoring

Meeting State

- Mute state
- Video state
- Raise-hand state
- Reaction state
- Pinned participant
- Active speaker
- Participant presence
- Screen-sharing state

Design System

- Primary Blue: **#0B5CFF**
- Secondary Blue: **#2D8CFF**
- Accent Yellow: **#FEBD00**
- Dark Meeting Background: **#0E0E0E – #1A1A1A**
- Light Dashboard Background: **#FFFFFF**
- Primary Text: **#232333**
- Typography: **Inter / Manrope / DM Sans**

The source specifies a consistent 8px spacing system, rounded 8–12px components, pill-shaped buttons and a shared design language across all meeting screens.

Architecture

User → ConnectMeet UI → Meeting Room → Real-Time Media Layer → Participants

For meeting actions:

Create / Join → Meeting Room → Audio + Video + Participant State → Real-Time Synchronization

05. CORE VIDEO CONFERENCING WORKFLOW

The main ConnectMeet workflow connects meeting creation, joining, device setup and live communication.

Meeting Workflow

Dashboard → New Meeting / Join → Pre-Join Lobby → Meeting Room → Controls → Leave / End Meeting

New Meeting

User selects **New Meeting**.

The system creates a unique meeting room and shareable meeting link/ID.

Join Meeting

User enters a meeting ID or link and display name.

The system validates the meeting ID before allowing the user to continue.

Pre-Join Lobby

The user sees a live camera preview and can configure microphone and camera settings before entering the meeting.

Live Meeting

- After joining, participants can:
- Mute / unmute
- Enable / disable camera
- Share screen
- View participants
- Raise hand
- Send reactions
- Chat
- Pin participants
- Switch between speaker and gallery layouts

Meeting Exit

- A participant can leave the meeting.
 - The host can end the meeting for all participants.
 - The application also cleans up the participant's media connection when leaving.
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06. FEATURE MAP

Home Dashboard

Provides access to the primary meeting actions:

- New Meeting
- Join
- Schedule
- Upcoming Meetings
- Past Meetings
- Profile
- Settings

Create / Join Meeting

- Users can switch between:
- Join Meeting
- New Meeting
- The join flow accepts a meeting ID or link and display name.
- The new meeting flow generates a unique meeting room.

Pre-Join Lobby

- Live camera preview
- Microphone toggle
- Camera toggle
- Display name
- Device permission state
- Join Meeting button
- Background blur / virtual background option

Video & Audio Calling

- Real-time video
- Real-time audio
- Participant video tiles
- Mute state synchronization
- Camera state synchronization
- Avatar fallback

In-Call Controls

- Mute / Unmute
- Camera On / Off
- Participants
- Chat
- Reactions
- Raise Hand
- Screen Share
- Host Tools
- More
- Leave / End Meeting

Gallery View

- Responsive video grid
- 4 participant layout
- 9 participant layout
- 16 participant layout
- Active speaker highlight
- Camera-off avatars

Speaker View

- Large active speaker video
- Participant name
- Microphone state
- Persistent control bar
- Meeting information

Screen Sharing

- Shared content as primary view
- Participant thumbnails
- Stop Share control
- Single active screen share

The screen-sharing specification states that shared content becomes the main canvas while participant videos collapse into a smaller thumbnail strip.

Participants

- Participant list
- Search
- Mic status
- Camera status
- Raise-hand indicator
- Pin participant
- Mute participant
- Mute all
- Remove participant
- Invite

Reactions

- Emoji picker
- Animated emoji reactions
- Raise-hand indicator
- Raised-hand queue

Active Speaker

Detects the current speaker based on audio input level.

The active speaker is visually highlighted and can be promoted to the main view. Debouncing is included to avoid rapid switching between speakers.

Auto Reconnect

- Detects unstable connections
 - Displays reconnecting status
 - Attempts automatic reconnection
 - Preserves meeting state
 - Removes participant only after reconnection attempts are exhausted
-

07. BUILD: DESIGN MODE

Objective

Create a complete UI screen set for **ConnectMeet**, a Zoom-style video conferencing web application MVP.

The design was created as a connected design system where all screens share the same colors, typography, spacing and component styles.

Design Direction

- Light theme for dashboard screens
- Dark theme for meeting screens
- Professional video conferencing interface
- Rounded components
- Pill-shaped buttons
- Clean video tiles
- Responsive layouts
- Minimal shadows
- Clear active states
- Consistent spacing

Screens Designed

1. Landing / Home Dashboard
2. Create / Join Meeting Modal
3. Pre-Join Lobby
4. In-Call Single Speaker View
5. In-Call Grid / Gallery View
6. Bottom Control Bar
7. Participants Panel
8. Screen Share Active State
9. Reactions & Raise Hand Overlay
10. Reconnecting / Connection Lost State
11. Responsive Mobile Layout

The Design Mode specification defines all 11 screens as part of one connected product system.

Visual System

Primary: #0B5CFF

Secondary: #2D8CFF

Accent: #FEBD00

Meeting Background: #0E0E0E – #1A1A1A

Dashboard: #FFFFFF

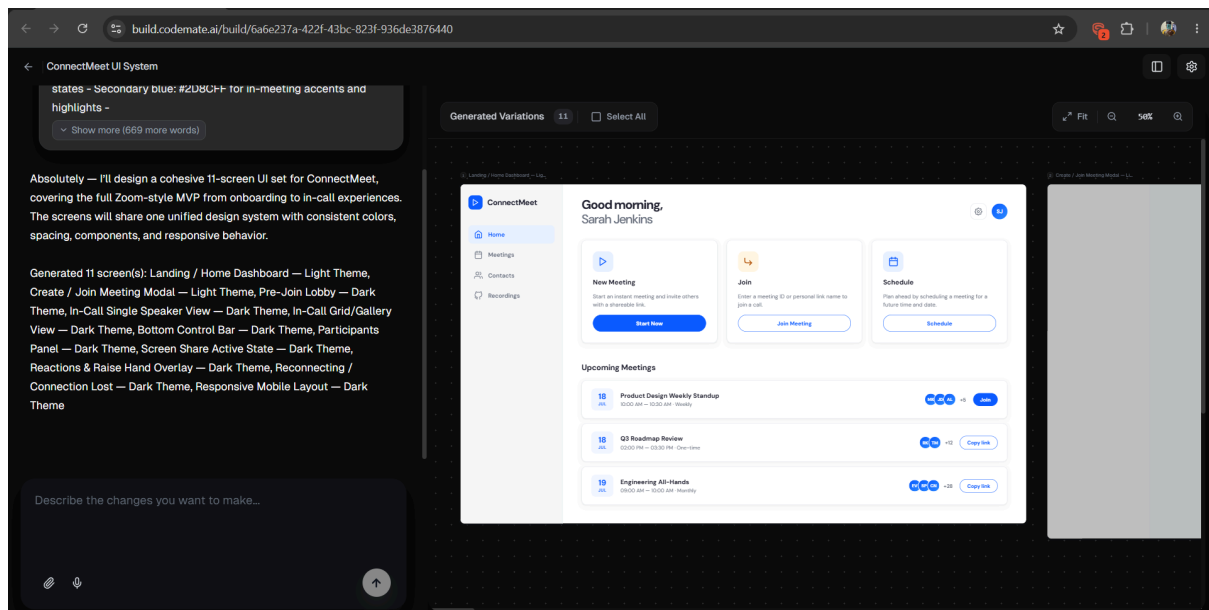
Text: #232333

Typography: Inter / Manrope / DM Sans

Spacing: 8px grid

Corners: 8–12px

Buttons: Rounded / pill-shaped



08. BUILD: PROTOTYPE MODE

Objective

Convert the approved ConnectMeet designs into a fully functional Zoom-style video conferencing MVP while maintaining the generated design system.

The Build prompt explicitly instructed the implementation to match the designs generated in Design Mode.

Implementation

- Build was used to create:
- Home Dashboard
- Meeting Creation
- Meeting Joining
- Meeting Scheduling
- Pre-Join Lobby
- Video Meeting
- Gallery View
- Speaker View
- Participant Panel
- Screen Sharing
- Reactions
- Raise Hand
- Meeting Controls
- Responsive Mobile Layout

Meeting Functionality

New Meeting

Creates a unique meeting room instantly.

Join Meeting

Allows users to enter a meeting ID or link.

Schedule

Opens the meeting scheduling flow.

Pre-Join

Provides live camera preview and device controls.

Live Meeting

Provides real-time video and audio communication.

Participant Management

Allows users and hosts to view and manage participants.

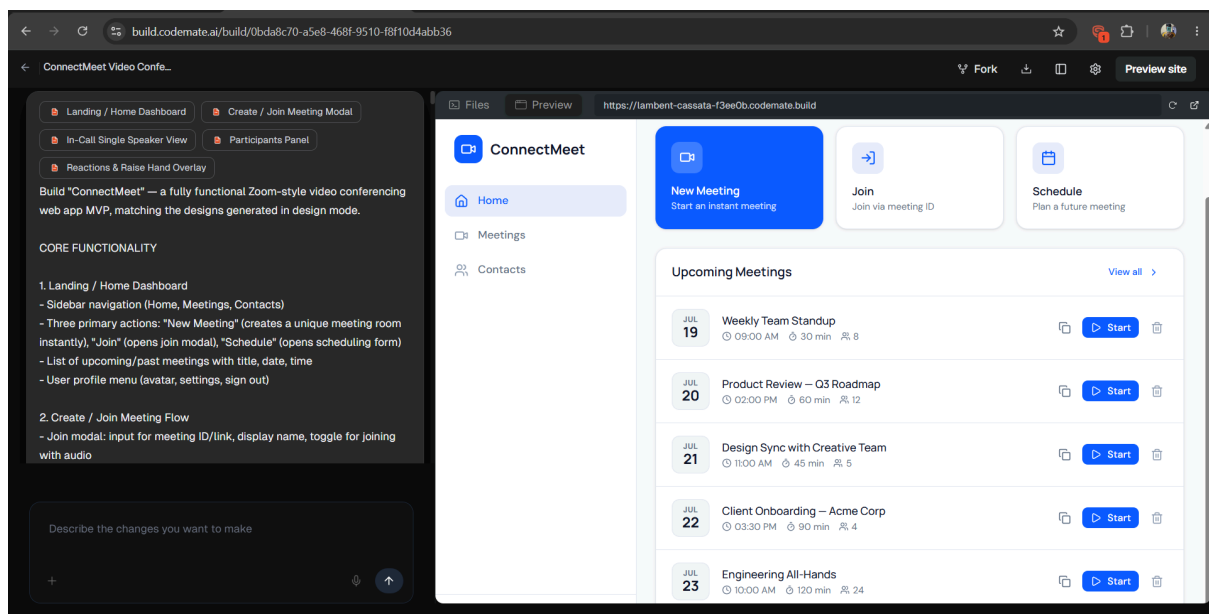
Screen Sharing

Allows participants to share their screen.

Responsive Layout

Automatically changes the video grid and controls based on available screen size.

The Build implementation requirements explicitly include real-time media, participant state synchronization, screen sharing, active speaker detection, pinning, reactions, reconnection and responsive behavior.



09. BUILD PROMPT HIGHLIGHTS

Design Mode

The main design prompt focused on building a complete Zoom-style conferencing interface rather than separate disconnected screens.

Prompt 1 : ConnectMeet UI Design

Design a complete UI screen set for "ConnectMeet" - a Zoom-style video conferencing web app MVP. Generate all screens as a cohesive design system using a professional blue-based visual identity, dark in-meeting interface, light dashboard interface, rounded components, pill-shaped buttons, clear active states and responsive layouts. Include dashboard, meeting creation, joining, pre-join lobby, speaker view, gallery view, participant panel, screen sharing, reactions, reconnecting state and mobile layouts.

Prompt 2 : Connected Design System

Generate all 11 screens as a connected design system sharing the same color tokens, typography scale, spacing and component styles so they feel like one cohesive product.

Prototype Mode

Prompt 3 : Functional ConnectMeet Application

Build "ConnectMeet" - a fully functional Zoom-style video conferencing web app MVP, matching the designs generated in Design Mode.

The prompt then defined the dashboard, meeting creation, joining, pre-join lobby, real-time video/audio, participant controls, screen sharing, active speaker detection, pinning, reactions, auto reconnect and responsive behavior.

Prompt 4 : Real-Time Meeting Functionality

Implement real-time peer-to-peer or SFU-based video and audio between meeting participants. Synchronize microphone, camera, hand-raise, pin and active-speaker states across clients while gracefully handling camera failures, permissions, participant joins, host changes and network interruptions.

Prompt 5 : Responsive Meeting Experience

Make the complete application responsive across desktop, tablet and mobile. Collapse the bottom control bar into icon-only controls on smaller screens, convert the participant panel into a mobile slide-up sheet and dynamically reflow the video grid based on screen size and participant count.

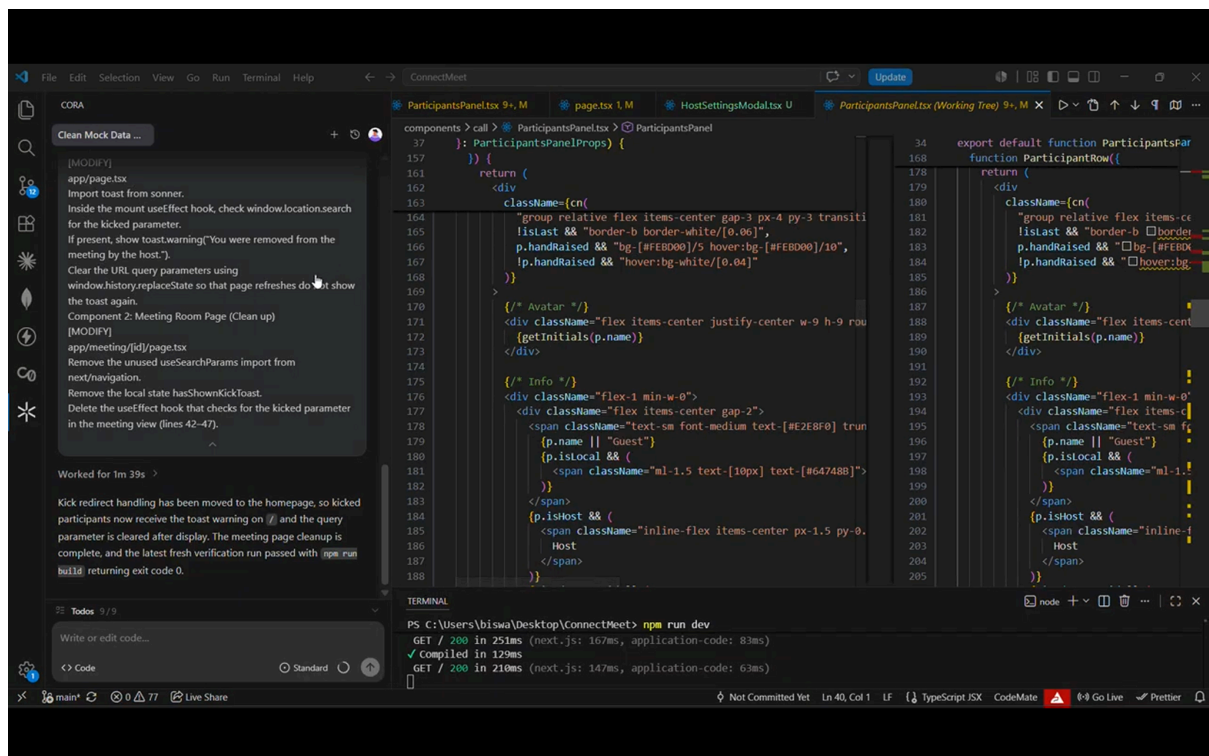
10. CORA: PLANNING & IMPLEMENTATION

Main Objective

Transfer the generated ConnectMeet project from Build into CORA for further implementation and development.

Development Flow

Design Mode → Build Code Mode → Functional MVP → Export ZIP → CORA



11. CORA TASK ARCHIVE

Project Handoff

The ConnectMeet project generated through Build was exported as a ZIP file for use with CORA.

CORA Documentation Status

- The supplied document does **not** contain:
- Authentication prompts
- Backend prompts
- Database prompts
- WebRTC implementation prompts
- CORA debugging logs
- CORA task completion records

Therefore, these are intentionally not claimed as completed implementation tasks.

12. IMPLEMENTATION MILESTONES

01. Design

Created the complete ConnectMeet conferencing UI using Build Design Mode.

02. Design System

Established shared colors, typography, spacing, buttons, video tiles, panels and meeting controls.

03. Frontend

Converted the approved designs into a functional video conferencing MVP using Build.

04. Meeting Creation

Implemented unique meeting room creation and shareable meeting IDs/links.

05. Meeting Joining

Implemented meeting ID/link validation and participant joining flow.

06. Pre-Join

Added camera preview, microphone controls, camera controls and display-name configuration.

07. Video Conferencing

Implemented the required real-time video and audio communication workflow.

08. Meeting Controls

Added mute, camera, participants, chat, reactions, screen sharing, host tools and leave/end controls.

09. Collaboration

Added screen sharing, participant management, active speaker detection, pinning, raise hand and reactions.

10. Reliability

Added reconnecting behavior and graceful handling of unstable connections.

11. Responsive UI

Implemented desktop, tablet and mobile meeting layouts.

12. CORA Handoff

Exported the generated project as a ZIP for further development in CORA.

13. BACKEND, DATA & REAL-TIME COMMUNICATION

ConnectMeet's primary technical requirement is real-time communication between participants.

Real-Time Media

The application supports:

- Video tracks
- Audio tracks
- Participant presence
- Mute state
- Camera state
- Screen sharing
- Connection status

Communication Model

The specification allows for either:

Peer-to-Peer communication

or

SFU-based communication

between participants in a meeting room.

Meeting State

- The system synchronizes:
- Mute state
- Video state
- Hand-raise state
- Pinned participant
- Active speaker
- Participant presence
- Screen-sharing state

The source specifically requires meeting state to remain synchronized across participants in real time.

Edge Cases

- No camera permission
 - No microphone permission
 - Participant joining mid-call
 - Host leaving
 - Network drop
 - Empty meeting rooms.
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14. AUTHENTICATION & USER MANAGEMENT

The supplied ConnectMeet document does not specify a particular authentication provider or detailed authentication implementation.

Therefore, no specific authentication technology is being claimed as implemented.

User-Level Features Documented

Profile

- User avatar
- Display name
- Profile menu
- Settings
- Sign out

Meeting Identity

Display name is collected during the join flow and can be edited before entering the meeting.

Host Controls

- Mute individual participants
- Mute all participants
- Remove participant
- End meeting for everyone
- Invite participants

Participant Controls

- Mute / Unmute
 - Camera On / Off
 - Share Screen
 - Leave Meeting
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15. RESPONSIVE UI & POLISH

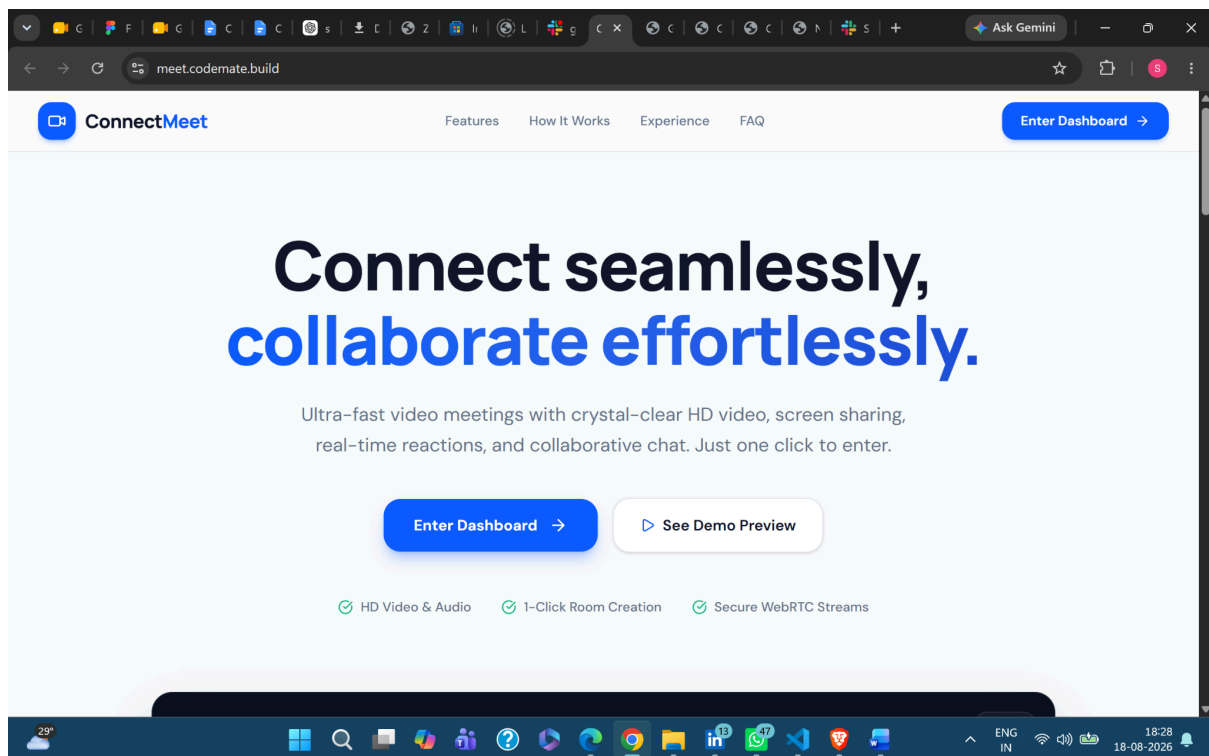
ConnectMeet was designed as a responsive video-conferencing application across desktop, tablet and mobile.

UI Characteristics

- Clean dashboard
- Dark meeting interface
- Blue primary accent
- Rounded video tiles
- Pill-shaped buttons
- Clear active/off states
- Consistent spacing
- Responsive video grid
- Touch-friendly controls
- Smooth meeting interactions

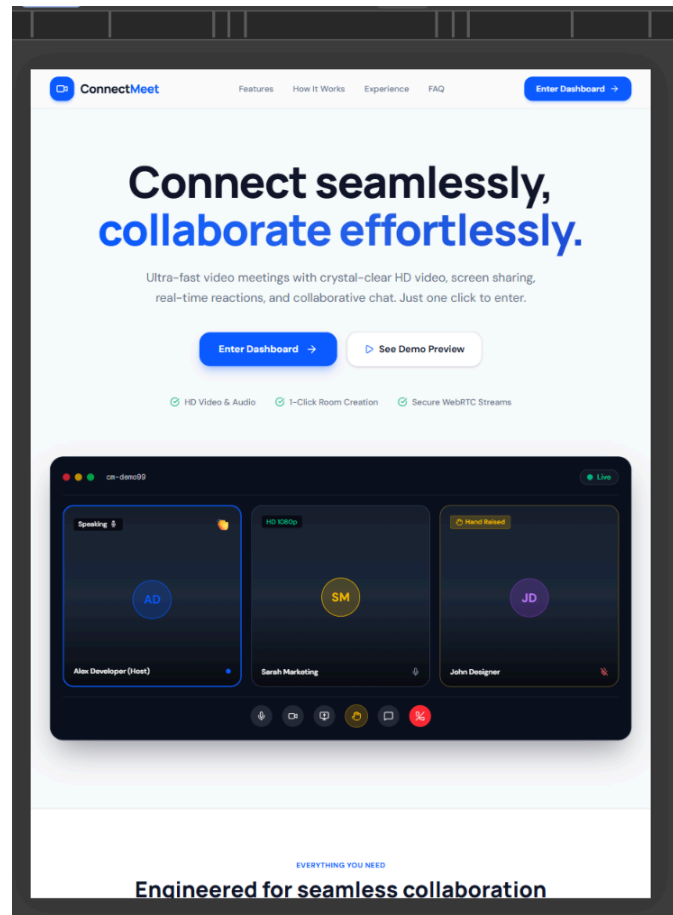
Desktop

- Full meeting control bar
- Large video canvas
- Gallery layouts
- Side participant panel
- Full meeting information



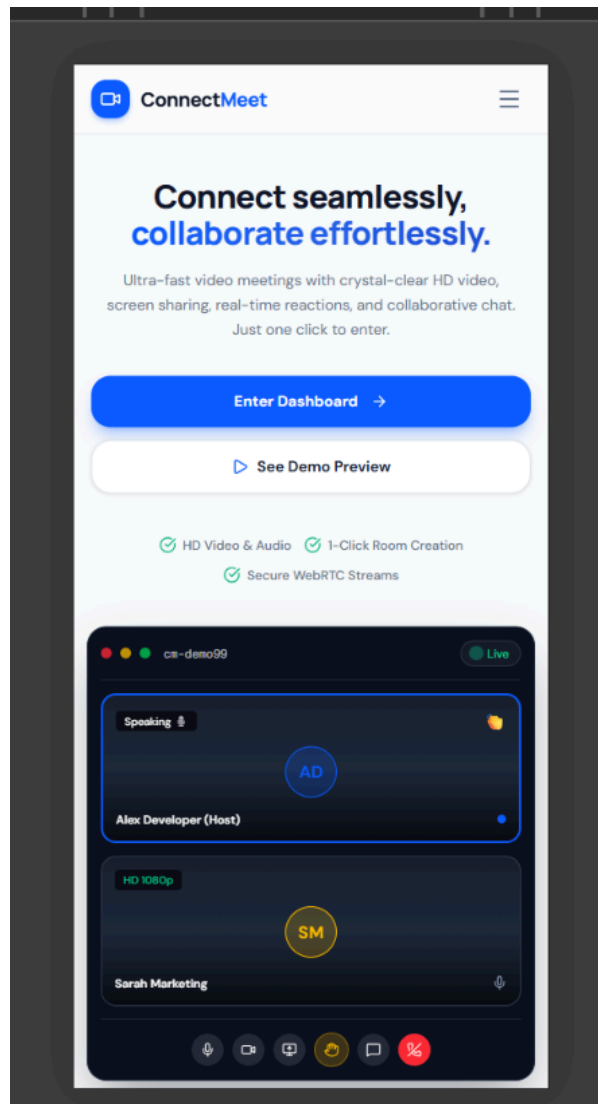
Tablet

- Adjusted video grid
- Responsive participant layout
- Condensed controls
- Optimized content density



Mobile

- Single-column video layout
- Icon-only bottom controls
- More overflow menu
- Full-screen participant sheet
- Dynamic video grid
- Portrait and landscape support



The mobile requirements specifically state that the control bar should collapse to icon-only buttons, the participant panel should become a full-screen slide-up sheet, and the video grid should automatically reflow based on screen width and participant count.

FINAL CONNECTMEET DEVELOPMENT FLOW

Design Mode



11-Screen Connected Design System



Build Code Mode



Functional Video Conferencing MVP



Meeting Creation + Joining



Real-Time Audio + Video



Screen Sharing + Participant Controls



Active Speaker + Pinning



Raise Hand + Reactions



Auto Reconnect



Responsive Mobile Experience



Export ZIP to CORA
